

Program: Diploma in Cloud Computing and Big data	
Course Code: 3274	Course Title: Digital Computer Architecture
Semester: 3	Credit:3
Course Category: Program Core	
Periods per week: 3 (L:3T:0 P:0)	Periods per semester: 45

Course Objectives:

- Understand the data representation in the Digital Computer System and Perform number conversion from one number system to another.
- Understand use of Boolean algebra and K-Map in digital circuits.
- Equip students to design and develop simple combinational and sequential circuits.
- Give an understanding about the basic architecture of 8085 and impart knowledge about assembly language programs of 8085

Course Prerequisites:

Topic	Course Code	Course name	Semester
To know semiconductor physics, diodes, Transistors.		Fundamentals of Electrical and Electronics Physics II	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Demonstrate data representation methods and binary arithmetic operations	8	Applying

CO2	Apply Boolean algebra concepts and realize Karnaugh Map for the logic functions.	9	Applying
CO3	Demonstrate the designing of combinational and sequential circuits	13	Applying
CO4	Illustrate assembly language programs in 8085 Microprocessor	13	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2	-	-	-	-	-
CO2	2	3	3	—	-	-	-
CO3	2	2	3	-	-	-	-
CO4	3	3	3	2	-	-	-

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain Data representation methods and binary arithmetic operations		
M1.01	Differentiate binary codes -binary, octal, hexadecimal, decimal and alphanumeric codes	3	Understanding
M1.02	Demonstrate the conversions from one number system to different number systems.	3	Applying
M1.03	Compute arithmetic operations such as addition and subtraction on binary numbers.	1	Applying

M1.04	Illustrate numbers in 1's and 2's complement and 2's complement arithmetic	1	Applying
Contents • Number systems and codes • Differentiate binary, Octal Hexadecimal and decimal number system • Codes: BCD, Gray, Excess-3, ASCII, EBCDIC. • Conversion from binary to octal, decimal and hexadecimal number systems and vice versa • Perform arithmetic operations such as binary addition and subtraction • Complements: 1's and 2's complements-2's complement arithmetic			
CO2	Apply Boolean algebra concepts and realize Karnaugh Map for the logic functions.		
M2.01	Explain basic theorems and postulates in boolean algebra	1	Understanding
M2.02	Demonstrate simplification of Boolean functions using theorems and postulates	1	Applying
M2.03	Derive the boolean function into canonical and standard forms such as Sum of Product (SOP) and Product Of Sum(POS)	1	Applying
M2.04	Describe functions of Logic gates	2	Understanding
M2.05	Implement Boolean functions using logic gates	1	Applying
M2.06	Use Kmap to simplify boolean expressions	3	Applying
	Series Test – I	1	
Contents • Boolean algebra- Basic theorems and properties. • Simplification of Boolean expression using theorems and properties • Boolean functions: Canonical and standard forms-Sum of Product (SOP) and Product of Sum (POS) expressions. • Basic Digital Logic Gates: Symbol, operation and truth-table of AND, OR, NOT, NAND, NOR, EX-OR, EX-NOR gates.and the concept of universal gates • Implement the Boolean functions using basic gates. • Simplify Boolean expression using K-map. • SOP and POS simplification-using upto Four variable Karnaugh map(K-map)			
CO3	Demonstrate the designing of combinational and sequential circuits		

M3.01	Describe combinational logic circuits	1	Understanding
M3.02	Demonstrate the design of combinational circuits	4	Applying
M3.03	Differentiate the operation of storage elements in sequential circuits such as latches and flip flops	3	Understanding
M3.04	Explain Data formats and different types of registers	1	Understanding
M3.05	Outline the working of various registers	4	Understanding
Contents: - Combinational logic circuits -Half adder, full adder, half Subtractor, full subtractor, Decoder Encoder, Multiplexer and Demultiplexer - Sequential logic circuits -Latch and Flip-flop,SR Latch,RS Latch- S-R flip-flop, D flip flop, J-K flip flop, JK master slave flip flop, T Flip Flop ,Synchronous and asynchronous sequential circuits - Registers: DataFormats- Classification of Registers-Shift Registers- SISO,SIPO,PISO,PIPO.			
CO4	Illustrate assembly language programs in 8085 Microprocessor		
M4.01	Describe microprocessor based computer system architecture	1	Understanding
M4.02	Explain the architecture of 8085	2	Understanding
M4.03	Summarize the 8085 Assembly Language programming	3	Understanding
M4.04	Demonstrate the 8085 instructions, addressing modes and assembly language programs	7	Applying
	Series Test – II	1	

Contents:

- Organization of microprocessor based system
- A Detailed Look at the 8085 MPU and its architecture-The 8085 microprocessor pinout and signals
- Introduction to 8085 Assembly Language Programming-8085 programming model-Instruction, Format and storage-Instruction word size, opcode format-How to write, assemble and execute simple program.
- Introduction to 8085 instructions-Data transfer(copy) operations, Addressing modes, Arithmetic operations, Logic operations, branch operations

Text / Reference

T/R	Book Title/Author
T1	M. Morris Mano & Michael D. Ciletti, Digital Design , Pearson Education, 5th Ed.
T2	A. Anand Kumar, Fundamentals of digital circuits , PHI Learning Pvt. Ltd., 2003
T3	Ramesh Gaonkar ,Microprocessor architecture ,Programming,and applications with 8085,5th Ed.
R1	Malvino & Leach, Digital Principles and Applications , McGraw-Hill,8 th Edition
R2	R P Jain, Modern Digital Electronics, McGraw-Hill,3 rd Edition

Online Resources

Sl.No	Website Link
1	http://www.asic-world.com/digital/tutorial.html
2	https://www.elprocus.com/8085-microprocessor-architecture/
3	https://www.geeksforgeeks.org/digital-electronics-logic-design-tutorials/
4	https://www.tutorialspoint.com/microprocessor/microprocessor_8085_architecture.htm
5	https://en.wikipedia.org/wiki/Digital_electronics